

Studying the role of potassium channels in mouse model of *Kcnv2* retinopathy.

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KCNV2 retinopathy, caused by mutations within the *KCVN2* gene, is a slow progressing inherited condition that leads to complete blindness. The *KCVN2* gene, which encodes for the voltage-gated potassium (Kv) subunit Kv8.2. Kv8.2 pairs with the Kv2.1 subunit to form a functional heterotetrameric potassium channel. The diagnosis of *KCNV2* retinopathy is confirmed through an electroretinogram (ERG), which measures the retina's electrical activity in response to light, as patients exhibit a unique ERG phenotype called a supernormal b-wave response. This study utilised wildtype (WT), Kv8.2 knockout (KO) and Kv2.1 KO mouse models to probe the origins of this supernormal b-wave. Three different pharmacological agents, L-AP4, BaCl₂, and Guangxitoxin-1E, were used to temporarily modulate the scotopic ERG response. The pharmacological agents were administered directly into the eye via an intravitreal injection to mice aged between 6-8 weeks. This was followed by ERG recordings 16-24h later. LAP-4 and BaCl₂ decreased b-wave amplitude in Kv2.1 KO and WT (only at lower light intensities). The application of guangxitoxin-1E did not affect WT and Kv2.1 KO retinas but lead to a decrease in ERG amplitude of Kv8.2 KO in both a- and b-wave suggesting that a negative voltage shift of the Kv2.1 homomeric channels lead to the supernormal b-wave. The results show that both bipolar cells and Muller glia contribute to the b-wave and the supernormal b-wave. Further exploration is needed into the contribution of homomeric Kv2.1 channels to this unique phenotype.